EXHIBIT A: IPv6 EXTENSION HEADER LAYOUT AND SMARTNIC VERIFICATION LOGIC

Centrifugal Compute Topology — In-Band Hardware Enforcement

1. IPv6 Option Header Layout (Capability Token)

The capability token is embedded into a custom IPv6 Destination Option Header ($0x3C$) processed in-band on TIP SmartNIC hardware at line-rate.

0	8	16	24 31
Next Header	Hdr Ext Len	Option Type ($0x1E$)	Opt Data Len (24)
Epoch Nonce (N_{epoch}) — 32 Bits			
PUF Key ID / Key Version (K_{puf_id}) — 32 Bits			
HMAC-SHA256 Capability Token (T_{cap}) [Bits 0–31]			
HMAC-SHA256 Capability Token (T_{cap}) [Bits 32–63]			
HMAC-SHA256 Capability Token (T_{cap}) [Bits 64–95]			
HMAC-SHA256 Capability Token (T_{cap}) [Bits 96–127]			

2. SmartNIC In-Band Pipeline Verilog Enforcement (Execution Module)

Listing 1: SmartNIC In-Band Interdiction Engine

```
module tip_inband_interdictor (
    input  wire         clk,
    input  wire         rst_n,
    input  wire [127:0] ipv6_hdr_token,
    input  wire [32:0]  ipv6_epoch_nonce,
    input  wire [255:0] local_puf_secret,
    output reg          action_drop
);
    wire [127:0] expected_hmac;

    // Local stateless HMAC verification derived from optical PUF key
    crypto_hmac_128 u_hmac (
        .clk(clk),
        .key(local_puf_secret),
        .msg({ipv6_epoch_nonce, 96'h0}),
        .digest(expected_hmac)
    );

    always @(posedge clk or negedge rst_n) begin
        if (!rst_n) begin
            action_drop <= 1'b0;
        end else begin
            // Statutory Blind Interdiction: Drop if Token matches & Epoch valid
            if (ipv6_hdr_token == expected_hmac) begin
                action_drop <= 1'b1; // Execute microsecond packet drop
            end else begin
                action_drop <= 1'b0; // Blind Common-Carrier Forward
            end
        end
    end
endmodule
```